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Idea Submission

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| Team Name | Debris |
| Team ID | 506 |
| Problem Statement Title | Debris Careers — Intelligent Career Guidance & Strea... |
| Theme / Bucket | Smart Assistants & Chatbots |
| Team Leader | Sarthak Soni, A26BSCAIU0010, A26BSCAIU0010@bennett.e... |

# Team Details



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# Problem Statement



## What is the problem?

- Millions face career ambiguity: Over 70% of high school and college students struggle with stream selection, leading to misplaced degrees and high dropout rates.
- Static advice & information overload: Conventional guidance relies on outdated articles that fail to understand a student's unique blend of skills, marks, and genuine passions.
- Scattered institutional intelligence: Cutoffs, premier colleges (IITs, AIIMS, NLUs, SRCC), and recruiter hiring criteria are fragmented across dozens of unverified portals.

## Who faces it?

- High School Students (Class 10th & 12th): Confused between Non-Medical (STEM), Medical, Commerce, and Humanities.
- Undergrads & Fresh Graduates: Seeking high-growth technical specializations, verified recruiter roadmaps, and salary tiers.
- Parents & Academic Mentors: Seeking unbiased, data-backed career guidance without predatory sales pitches from commercial coaching institutes.

## Why do current options fall short?

- Private counsellors are costly & subjective: 1-on-1 human sessions charge ₹5,000–₹25,000 and often suffer from individual bias, affiliate marketing tie-ins, and lack of real-time employment data.
- Static tests & MCQs are rigid: Conventional aptitude questionnaires force students into artificial multiple-choice boxes rather than parsing free-form natural language expressions of their personal interests.
- Generic AI chatbots hallucinate: General-purpose LLMs frequently invent fake cutoffs, non-existent entrance requirements, and unverified placement statistics without strict deterministic grounding.

# Proposed Solution



## Our idea in one line

- We are building Debris Careers, an intelligent NLP and Machine Learning powered Career Advisory assistant that transforms free-form natural language queries into deterministic, ranked career roadmaps, college benchmarks, and recruiter intelligence without forced MCQs.

## How it solves the problem

- Free-Form Natural Intent Understanding: Students freely describe their passions, favorite subjects, or career goals in natural conversational language.
- Vector-Space ML Matching: TF-IDF vectorization and cosine similarity score user input against verified institutional career profiles in real-time.
- Complete Actionable Roadmaps: Synthesizes required academic degrees, top-tier colleges, entrance exams, skills, hiring companies, and sequential next steps.

## What makes it different

- Zero Hallucination Guarantee: Combines machine learning ranking with a deterministic knowledge base, ensuring every recommended college and exam is 100% verified.
- Context-Aware Follow-ups: Built-in conversation memory remembers previously discussed careers to seamlessly answer targeted follow-ups (colleges, skills, exams).
- Privacy-First & Fully Offline Capable: Zero reliance on paid external cloud APIs; runs locally with SQLite session persistence and instant chat history management.

# Technical Approach



## Planned tech stack

- **Python 3**
- **Flask Framework**
- **Scikit-Learn (TF-IDF)**
- **SQLite3 Database**

## How the system will work

**Student asks query or inputs profile**

Free-form natural text (e.g., 'coding, AI, math, high paying jobs')

→

**NLP preprocessor extracts intent**

Tokenizes input, detects academic stream and core interest keywords

→

**TF-IDF Vectorizer & Cosine Similarity**

Computes mathematical similarity against institutional career vector space

→

**Advisory delivered & stored in SQLite**

Synthesizes ranked colleges, exams, and recruiters; persists conversation context

## What we have started working on

- Implemented & Benchmarked Core ML Pipeline: Built the Scikit-Learn TF-IDF vectorization and cosine similarity scoring algorithm across 4 primary streams (STEM, Medicine, Commerce, Humanities) with 100% deterministic ranking accuracy.
- Engineered Full-Stack Flask Web Application: Built secure salted SHA-256 user authentication, SQLite conversation state persistence, and real-time conversation deletion and history clearing endpoints.
- Crafted Responsive Lavender Glassmorphism UI: Implemented a minimalist, premium frontend with auto-expanding input, time-aware greetings, message timestamps, thinking animation, and collapsible mobile drawer.

# Impact and Next Steps



## ▲ Expected impact

- Democratizes Career Guidance: Provides high-quality, unbiased career and college counselling to thousands of students completely free of charge.
- Reduces Course Dropouts: Accurately aligns genuine student aptitude and interests with appropriate degrees before costly college enrollment mistakes.
- Saves Weeks of Manual Research: Collates entrance exams, college rankings, skills, and recruiters into structured cards.

## ✓ Feasibility

- 100% Working Prototype Already Built: Backend server, machine learning pipeline, database, and client interface are fully operational and tested today.
- Zero API Costs & Hardware Agnostic: Requires no paid cloud tokens or heavy GPUs; operates fast and efficiently on standard laptop CPU hardware.
- Balanced 3-Member Team: Sarthak Soni (Team Lead & ML/Backend), Navneet Kumar (UI/UX & Research), Bhawin Sawant (Dev & Testing).

## ▶ Plan for the next round

- Phase 1: Expand dataset to 100+ specialized vocational and emerging tech pathways (Quantum, Bio-Robotics, Fintech).
- Phase 2: Integrate student transcript and resume parser to suggest customized career pivots based on existing coursework.
- Phase 3: Deploy cloud microservice with multi-region CDN and present live interactive demonstration to hackathon judges.

## § References and data sources


- Ministry of Education (NIRF India Rankings 2024–2025): Official institutional accreditation, cutoffs, and college tiers.
- National Career Service (NCS) & NASSCOM FutureSkills: Skill demand matrix and corporate compensation benchmarks.
- Scikit-Learn Information Retrieval Architecture: Vector Space Models & Cosine Similarity for deterministic matching.